Reusable adaptive source and incremental extension

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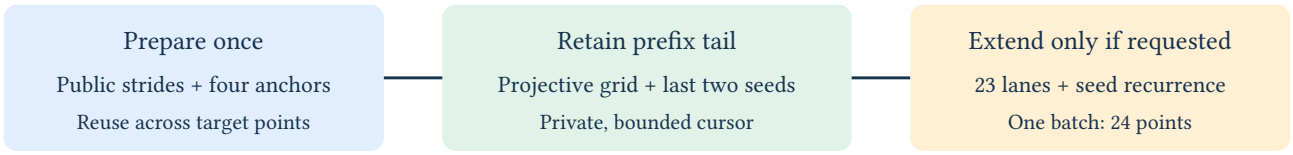
Prepared public source state is now retained across recoveries. A target-specific cursor keeps the final projective grid and the last two homogeneous seed pairs, so an exceptional character can extend the prefix. The adaptive coefficients and sign predicates are unchanged. [pcc:unverified]



Eight saved recoveries: four skip extension, four use recurrence.

Fixed F509 / r479 records in recovery.tsv; counts are not elapsed-time measurements.

State invariant

At prefix endpoint T, lane j represents [lambda_j]T. Add P to T and the cached affine [lambda_j]P to each lane, then batch-normalize the new point and lanes together. For seed pairs c(T)=N/D and c(T-P)=A/B, the next pair is [delta S N^2 B : D^2 A], with delta=x(P)-x(T) and S=c(P)^2. Anchors O, P, -P, 2P take precedence. A one-seed prefix can still require direct seeding. [pcc:unverified]

Cursor fields are private and lifetime-bound to preparation. Advance consumes the previous cursor; an error returns no replacement. An explicit remaining-step allowance is checked before advancing geometry. Identity steps skip normalization, while invalid lane configurations are rejected.

Recorded checks

The optimized overflow-checked suite reports89 passing tests. All320 split-window extensions match uninterrupted signs, seed pairs, points and methods. A test-only counter checks actual calls to phase(). Another44 extensions on the F5/r7 and F7/r13 fixtures traverse repeated identity crossings:20 anchors and24 recurrences, with no additional direct phases. [pcc:unverified]

One prepared F509/r479 source serves all eight saved adaptive targets. Four require an extra sign; each uses recurrence,23 lane updates and one batch conversion of24 points. The split-window checks cover these same exceptional offsets without direct phase calls. A separate checker reexecutes the frozen pre-cursor source, accepts the eight outputs and query decisions, and rejects a corrupted recovered scalar. Its metadata checks do not prove producer execution. [pcc:unverified]

Cost and limits

Seed recurrence adds no field inversion; nonidentity geometry extension uses the existing single-inverse batch routine. Initial grid construction, direct initial phases, value/division and polynomial work remain. The prefix API still returns the complete prefix; only cursor retention is compact. No whole-source timing or inversion-free claim is made. [pcc:unverified]

Twenty-three source/data hashes bind the run; eleven compiler warnings are retained. The original4563-case cohort is preserved. No cloud job or timing panel ran. Next bind complete-cost instrumentation and investigate shorter adaptive prefixes with representable collapsed targets.

Evidence and prior work

Claims PCC-20261105-99d02c and PCC-20261105-cc8f6a await signed coverage. See DERIVATION.md, source-lock.json, origin-lock.json, receipts and raw logs. Source formulas follow the preserved shared-seed and pair-seed derivations. Jacobian conventions were checked against the Explicit-Formulas Database. No novelty claim.